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3D integrated circuit (3DIC) structure

Abstract

An embodiment bonded integrated circuit (IC) structure includes a first IC structure and a second IC structure bonded to the first IC structure. The first IC structure includes a first bonding layer and a connector. The second IC structure includes a second bonding layer bonded to and contacting the first bonding layer and a contact pad in the second bonding layer. The connector extends past an interface between the first bonding layer and the second bonding layer, and the contact pad contacts a lateral surface and a sidewall of the connector.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6326701	12/2000	Shinogi et al.	N/A	N/A
7242099	12/2006	Lin	257/E23.079	H01L 24/73
7385283	12/2007	Wu et al.	N/A	N/A
7564115	12/2008	Chen et al.	N/A	N/A
7633165	12/2008	Hsu et al.	N/A	N/A
7825024	12/2009	Lin et al.	N/A	N/A
7973413	12/2010	Kuo et al.	N/A	N/A
8105875	12/2011	Hu et al.	N/A	N/A
8158456	12/2011	Chen et al.	N/A	N/A
8183578	12/2011	Wang	N/A	N/A
8183579	12/2011	Wang	N/A	N/A
8227902	12/2011	Kuo	N/A	N/A
8278152	12/2011	Liu et al.	N/A	N/A
8426961	12/2012	Shih et al.	N/A	N/A

8669174	12/2013	Wu et al.	N/A	N/A
8802504	12/2013	Hou et al.	N/A	N/A
8803292	12/2013	Chen et al.	N/A	N/A
8803316	12/2013	Lin et al.	N/A	N/A
8860229	12/2013	Lin	257/774	H01L 24/02
10157889	12/2017	Chen	N/A	H01L 25/0657
10319701	12/2018	Yu	N/A	H01L 25/0657
2001/0050432	12/2000	Seyyedy	257/E23.125	H01L 23/49811
2004/0097002	12/2003	Pogge et al.	N/A	N/A
2006/0121690	12/2005	Pogge et al.	N/A	N/A
2007/0037379	12/2006	Enquist et al.	N/A	N/A
2009/0263951	12/2008	Shibata et al.	N/A	N/A
2009/0273914	12/2008	Supriya et al.	N/A	N/A
2010/0264551	12/2009	Farooq et al.	N/A	N/A
2012/0068355	12/2011	Aoki et al.	N/A	N/A
2012/0153484	12/2011	Sadaka	438/455	H01L 21/76856
2012/0228713	12/2011	Chen et al.	N/A	N/A
2013/0270700	12/2012	Yu	257/738	B23K 35/262
2014/0001645	12/2013	Lin et al.	N/A	N/A
2014/0225258	12/2013	Chiu	257/737	H01L 21/6835
2014/0252572	12/2013	Hou	438/112	H01L 25/0655
2014/0252632	12/2013	Barth	257/773	H01L 23/5384
2016/0197055	12/2015	Yu	257/737	H01L 24/81
2018/0301430	12/2017	Kuo	N/A	H01L 21/6835
2019/0123017	12/2018	Lu	N/A	H01L 24/17

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Background/Summary

PRIORITY CLAIM AND CROSS-REFERENCE (1) This application is a continuation of U.S. patent application Ser. No. 17/649,381, filed Jan. 31, 2022, entitled “3D Integrated Circuit (3DIC) Structure,” which is a divisional of U.S. patent application Ser. No. 16/435,697, filed Jun. 10, 2019, now U.S. Pat. No. 11,239,201, issued Feb. 1, 2022, entitled “3D Integrated Circuit (3DIC) Structure,” which is a divisional of U.S. patent application Ser. No. 14/591,784, filed Jan. 7, 2015, now U.S. Pat. No. 10,319,701, issued Jun. 11, 2019, entitled “3D Integrated Circuit (3DIC) Structure and Method of Making Same,” each application is hereby incorporated by reference in its entirety.

BACKGROUND

(1) The semiconductor industry has experienced rapid growth due to continuous improvements in the integration density of a variety of electronic components (e.g., transistors, diodes, resistors, capacitors, etc.). For the most part, this improvement in integration density has come from repeated reductions in minimum feature size (e.g., shrinking the semiconductor process node towards the sub-20 nm node), which allows more components to be integrated into a given area. As the demand for miniaturization, higher speed and greater bandwidth, as well as lower power consumption and latency has grown recently, there has grown a need for smaller and more creative packaging

techniques of semiconductor dies.

(2) As semiconductor technologies further advance, stacked semiconductor devices, e.g., 3D integrated circuits (3DIC), have emerged as an effective alternative to further reduce the physical size of a semiconductor device. In a stacked semiconductor device, active circuits such as logic, memory, processor circuits and the like are fabricated on different semiconductor wafers. Two or more semiconductor wafers may be installed on top of one another to further reduce the form factor of the semiconductor device.

(3) Two semiconductor wafers or dies may be bonded together through suitable bonding techniques. An electrical connection may be provided between the stacked semiconductor wafers. The stacked semiconductor devices may provide a higher density with smaller form factors and allow for increased performance and lower power consumption.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIGS. 1A-1B, 2A-2B, 3A-3B, 4A-4B, 5A-5B, and 6-9 illustrate various intermediary stages of manufacturing a bonded integrated circuit structure in accordance with some embodiments.

(3) FIG. 10 illustrates a process flow for forming the bonded integrated circuit in accordance with some embodiments.

DETAILED DESCRIPTION

(4) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(5) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(6) Various embodiments include a bonded integrated circuit structure, which may include a first integrated circuit structure bonded to a second integrated circuit structure. The bonding of the integrated circuit structures may include a hybrid bonding process where both bonding layers (e.g., oxide-to-oxide bonding layers) and conductive interconnect structures are used. The conductive interconnect structures may include a concave contact pad (e.g., having a U-shape in a cross-sectional view), which may be used to contain the material of a connector (e.g., solder or copper bump) during bonding. In the bonded structure, the concave contact pad may contact a lateral

surface and sidewalls of the connector. Thus, the risk manufacturing defects, such as, solder bridging may be reduced, which allows for conductive interconnect structures to be spaced closer together (e.g., have a smaller pitch), increasing I/O density and improving yield.

(7) FIGS. 1A through 5B illustrate the formation of various structures for bonding two integrated circuit structures **100** and **150** prior to bonding. Although the figures illustrate an embodiment of a die-on-wafer bonding process, other embodiments may also be applied to die-on-die bonding, wafer-on-wafer bonding, and the like. FIGS. 1A and 1B illustrate a die **100** (see FIG. 1A) and a wafer **150** (see FIG. 1B) prior to bonding. In some embodiments, die **100** may be a semiconductor die and could be any type of integrated circuit, such as a processor, logic circuitry, memory, analog circuit, digital circuit, mixed signal, and the like. Similarly, wafer **150** may include any type of integrated circuits, such as a processor, logic circuitry, memory, analog circuit, digital circuit, mixed signal, and the like. The functionality provided by die **100** and wafer **150** may or may not be the same.

(8) Throughout the FIGS. 1A through 5B, Figures ending in designation “A” illustrate die **100** and figures ending in designation “B” illustrate wafer **150** prior to bonding. Furthermore, although the figures and description describe the parallel formation of interconnect structures on die **100** and wafer **150**, other embodiments may apply to the formation of such structures at different times. For example, interconnect structures (e.g., contacts **114**, see FIG. 5A) in die **100** may be formed prior to, simultaneously, and/or after interconnect structures (e.g., contact pads **164**, see FIG. 5B) in wafer **150**. Although described as a die **100** throughout, one of ordinary skill will readily understand that some processing on die **100** may occur while die **100** is part of a larger substrate, for example, a wafer.

(9) Die **100** and wafer **150** include a substrate **102** and a substrate **152**, respectively. Substrates **102** and **152** may comprise, for example, bulk semiconductor, doped or undoped, or an active layer of a semiconductor-on-insulator (SOI) substrate. Generally, an SOI substrate comprises a layer of a semiconductor material, such as silicon, formed on an insulator layer. The insulator layer may be, for example, a buried oxide (BOX) layer or a silicon oxide layer. The insulator layer is provided on a substrate, such as a silicon or glass substrate. Substrates **102** and/or **152** may include elementary semiconductor, such as silicon or germanium; a compound semiconductor including silicon carbide, gallium arsenic, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP; or combinations thereof. Other substrates, such as multi-layered or gradient substrates, may also be used. Active devices such as transistors, capacitors, resistors, diodes, photo-diodes, fuses, and the like may be formed at the top surface of each substrate **102** and **152**.

(10) Interconnect layers **104** and **154** may be formed over the active devices and substrates **102** and **152**, respectively. Interconnect layers **104** and **154** may include inter-layer dielectric (ILD) and/or inter-metal dielectric (IMD) layers containing conductive features **106** and **156** (e.g., conductive lines and vias comprising copper, aluminum, tungsten, combinations thereof, and the like), respectively, formed using any suitable method. The ILD and IMDs may include low-k dielectric materials having k values, for example, lower than about, 4.0 or even 2.0, disposed between such conductive features. In some embodiments, the ILD and IMDs may be made of, for example, phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), fluorosilicate glass (FSG), SiO.sub.xC.sub.y, Spin-On-Glass, Spin-On-Polymers, silicon carbon material, compounds thereof, composites thereof, combinations thereof, or the like, formed by any suitable method, such as spinning, chemical vapor deposition (CVD), and plasma-enhanced CVD (PECVD). Interconnect layers **104** and **154** electrically connect various active devices to form functional circuits within die **100** and wafer **150**, respectively. The functions provided by such circuits may include memory structures, processing structures, sensors, amplifiers, power distribution, input/output circuitry, or the like. The functions provided by circuits in die **100** and wafer **150** may or may not be the same. The above examples are provided for illustrative purposes only to further explain applications of

various embodiments. Other circuitry may be used as appropriate for a given application.

(11) As further illustrated by FIGS. 1A and 1B, interconnect layer **104** of die **100** and interconnect layer **154** if wafer **150** may further include a top-most interconnect layer having conductive features **106'** and **156'**, respectively. Conductive features **106'** may be a metal line, contact pad, or the like disposed on a top surface of die **104**. Similarly, conductive feature **156'** may be a metal line, contact pad, or the like disposed on a top surface of wafer **150**. The various features of die **100** and wafer **150** may be formed by any suitable method. Furthermore, the general features and configuration of die **100** and/or wafer **150** described above are but one example embodiment, and die **100** and/or wafer **150** may include any combination of any number of the above features as well as other features.

(12) Referring next to FIGS. 2A and 2B, bonding layers **108** and **158** may be disposed on a top surface of die **100** and wafer **150**, respectively. Bonding layer **108** may cover conductive feature **106'**, and bonding layer **158** may cover conductive feature **156'**. In some embodiments, bonding layers **108** and **158** may include an oxide (e.g., silicon oxide, silicon oxynitride, and the like), which may be formed using a suitable deposition process, such as spinning, CVD, atomic layer deposition (ALD), plasma enhanced CVD, physical vapor deposition (PVD), and the like. In subsequent process steps, bonding layers **108** and **158** may be used to bond die **100** to wafer **150** in a hybrid bonding process, for example. In some embodiments, bonding layers **108** and **158** may further function as a passivation layer.

(13) Subsequently, as illustrated in FIGS. 3A and 3B, bonding layers **108** and **158** are patterned to include openings **110** and **160** exposing conductive features **106'** and **156'**, respectively. The patterning of bonding layers **108** and **158** may include photolithography and etching processes. For example, photoresists (not shown) may be blanket deposited over bonding layers **108** and **158**, and the photoresists may be exposed, e.g., to light, using a photomask. Exposed or unexposed portions of the photoresists may then be removed depending on whether a positive or negative resist is used. Thus, the photoresists may be patterned to include openings, which may correspond to openings **110** and **160**.

(14) Subsequently, the pattern of the photoresists may be transferred to bonding layers **108** and **158** using etching processes. Thus, openings **110** exposing conductive features **106'** may be patterned in bonding layer **108**, and openings **160** exposing conductive features **156'** may be patterned in bonding layer **158**. The etching may be any acceptable etch process, such as a reactive ion etch (RIE), neutral beam etch (NBE), the like, or a combination thereof. The etching may be anisotropic. Subsequently, the photoresists are removed in an ashing and/or wet strip process(es), for example. In some embodiments, hard masks (not shown) might be formed between the photoresists and bonding layers **108** and **158**, in which embodiments the pattern from the photoresists would first be imposed upon the hard masks and the patterned hard masks would be used in patterning the underlying layers **108** and/or **158**.

(15) FIG. 4A illustrates the formation of a seed layer **112** over die **100** using any suitable method, such as sputtering, CVD, PVD, electroless plating, and the like. In an embodiment, seed layer **112** is a conformal layer. The formation of seed layer **112** may cover a top surface of bonding layer **108** in die **100**. In some embodiments, seed layer **112** may comprise a conductive material, such as, solder, copper, alloys thereof, combinations thereof, and the like, for example. Seed layer **112** may be disposed in openings **110** and may contact underlying conductive features **106'**. Conductive features **106'** may electrically connect seed layer **112** to the circuits in die **100**.

(16) Similarly, FIG. 4B illustrates the formation of a conductive layer **162** over a top surface of wafer **150** using any suitable blanket deposition process, such as, sputtering, CVD, PVD, electroless plating, and the like. In some embodiments, conductive layer **162** comprises copper, nickel, gold, tin, silver, aluminum, alloys thereof, combinations thereof, and the like, for example. Conductive layer **162** may be a conformal layer disposed on sidewalls and a bottom surface of openings **160**. Thus, conductive layer **162** may contact underlying conductive feature **156'**, which

may electrically connect conductive layer **162** to the circuits in wafer **150**. In some embodiments, conductive layer **162** may have a thickness **T1** of about 500 Å to about 8000 Å, for example. As illustrated by FIG. 4B, conductive layer **162** may cover a top surface of bonding layer **158** in wafer **150**.

(17) FIGS. 5A and 5B illustrate the formation of interconnect features in die **100** and wafer **150**. FIG. 5A illustrates the formation of convex connectors **114** (e.g., a conductive pillar or bump) by applying a plating process (e.g., electroless plating, electrochemical plating, and the like) on seed layer **112**. During the plating process, a mask (e.g., a photoresist or hard mask, not shown) may be disposed over die **100** to define a shape of connectors **114**. The plating process may consume seed layer **112** in openings **110** and thus, seed layer **112** is not separately illustrated in FIG. 5A. The resulting connectors may fill openings **110** and extend past a top surface of die **100**. After forming connectors **114**, excess portions of seed layer **112** (e.g., outside of openings **110**) may be removed using a suitable process, such as, photolithography and/or etching, for example. In some embodiments, connectors **114** may comprise solder, copper, combinations thereof, and the like.

(18) FIG. 5B illustrates the removal of portions of conductive layer **162** over bonding layer **158** using any suitable planarization process. For example, a chemical mechanical polish (CMP), grinding, or other suitable process may be used. Because conductive layer **162** covers a top surface of wafer **150**, defects caused by the planarization process (e.g., CMP) due to differences in pattern density, material mismatch, and the like may be reduced.

(19) In the resulting structure, remaining portions of conductive layer **162** in openings **160** forms contact pads **164**, which may be concave in configuration. For example, in the illustrated cross-sectional view, contact pads **164** have a U-shape, and in a top-down view (not shown), wafer **150** includes contact pads **164** surrounding openings **160**, which may be circular, elliptical, or the like in shape. Contact pads **164** may be disposed on a sidewall and bottom surface of openings **160**, and contact pads **164** may be electrically connected to underlying conductive features **156'**. In various embodiments, contact pads **164** may not completely fill openings **160**, and openings **160** may only be partially filled. For example, after the formation of contact pads **164**, openings **160** may have a lateral dimension **L1** of about 0.5 μm to about 20 μm and a vertical dimension **H1** of about 2 μm or less. As illustrated, lateral dimension **L1** may be defined as the horizontal dimension of openings **160** between inner sidewalls of a contact pad **164**. Furthermore, vertical dimension **V1** may be defined as a vertical dimension between an exposed, top surface of contact pad **164** to a top surface of bonding layer **158**. Throughout the description, the terms “horizontal” or “lateral” are defined as a direction parallel to a major surface (e.g., a surface having active devices) of substrates **102/152** whereas “vertical” is defined as a direction perpendicular to the major surface of substrates **102/152**. The terms “horizontal”, “lateral”, and “vertical” are relative and not meant to impose any sort of absolute orientation.

(20) In subsequent process steps, connector **114** may be disposed in remaining portions of opening **160** during the bonding of die **100** to wafer **150**. Thus, the dimensions of connector **114** and opening **160** may be related. For example, a portion of connector **114** extending past bonding layer **108** (labeled portion **114'**) may have an area of about 95% to about 100% of the area of opening **160**. Furthermore, in order to allow connector **114** to be disposed within opening **160**, a lateral dimension **L2** of connector **114** may be less than lateral dimension **L1** of opening **160**. In some embodiments, lateral dimension **L2** may be about 60% to about 80% of lateral dimension **L1**. For example, in embodiments where lateral dimension **L1** is about 0.5 μm, lateral dimension **L2** may be about 0.4 μm or less. As another example, in embodiments where lateral dimension **L1** is about 3.5 μm, lateral dimension **L2** may be about 2 μm or less. In some embodiments, the relative value of lateral dimension **L1** in relation to lateral dimension **L2** may depend on the bonding accuracy for bonding die **100** to **150**. For example, when higher bonding accuracy can be achieved, a ratio of lateral dimension **L2** to lateral dimension **L1** may be increased.

(21) Due to the configuration of contact pads **164** and bonding layer **158**, contact pads **164** may be

used to contain the material (e.g., solder) of connector **114** during the bonding of die **100** to wafer **150**. Thus, connectors **114** and contact pads **164** may be spaced relatively close together without the risk of connector bridging (e.g., solder bridging). For example, a pitch P1 between neighboring contact pads **164** may be less than about 10 μm or even less than about 5 μm . In embodiments where features of die **100** is part of a larger substrate (e.g., a wafer), a singulation process may be applied to separate die **100** from other features (e.g., other dies) in the substrate.

(22) FIGS. **6** and **7** illustrate the bonding of die **100** and wafer **150** using a hybrid bonding process, for example. In FIG. **6**, a pre-bonding process is illustrated. Die **100** and wafer **150** may be orientated so that connectors **114** and respective contact pads **164** face each other. For example, in the illustrated embodiment, die **100** may be disposed so that bonding layer **108** is orientated downwards. During pre-bonding, die **150** may be heated to a pre-bonding temperature less than the melting temperature of connectors **114**. For example, die **150** may be heated to a pre-bonding temperature of about 150° Celsius (C) to about 200° C. Die **100** may then be contacted to wafer **150** so that bonding layers **108** and **158** are in physical contact as indicated by arrows **168**. By heating die **150**, hydrogen bonds may be formed between bonding layers **108** and **158** when die **100** is contacted to wafer **150**. Furthermore, during pre-bonding connectors **114** may be aligned with openings **160** so that connectors **114** are disposed in openings **160** after bonding (see e.g., FIG. **7**).

(23) Subsequently, in FIG. **7**, an annealing process is applied bonding die **100** to wafer **150**, thus forming bonded integrated circuit structure **170**. In some embodiments, the annealing process may be conducted at a suitably high temperature to melt and bond connectors **114** to contact pads **164**. The annealing process may further form covalent bonds between bonding layers **108** and **158** in a bonding process. For example, the annealing process may include heating die **100** and wafer **150** to a temperature of about 250° C. to about 300° C. and maintaining this temperature for about an hour. Connectors **114** may extend past an interface between die **100** and wafer **150**. As discussed above, the shape of contact pads **164** may be used to contain the material of connectors **114** during bonding from undesired lateral spreading. For example, in the bonded integrated circuit structure **170**, contact pads **164** may be disposed on a lateral (e.g., bottom) surface and sidewalls of connectors **114**. Furthermore, the material of connectors **114** may not spread between bonding layers **108** and **158**. Thus, the risk manufacturing defects (e.g., bridging) may be reduced, allowing for interconnect structures having a smaller pitch between connectors **114** and contact pads **164**. Bonded connectors **114** and contact pads **164** may electrically connect circuits in die **100** to circuits in wafer **150**.

(24) In the resulting structure, portions (e.g., sidewall portions) of contact pads **164** may contact bonding layer **108**. However, due to the annealing process, the conductive material of connectors **114** may suffer shrinkage, which may result in the presence of air gaps **172** at the interface between die **100** and wafer **150**. Such air gaps **172** may be disposed between bonding layer **108** and contact pads **164**/portions of connectors **114** in openings **160**. For example, in some embodiments, a volume of the material of connectors **114** may shrink by about 5% to about 6%, and the volume of air gaps **172** may be about 5% to about 10% of the volume of connector **114** in openings **160**. The presence of air gaps **172** may advantageously be used for stress relaxation at the interface between die **100** and wafer **150** as well as to reduce air trapping within the conductive material of connector **114**. Thus, in some embodiments, the volume of connectors **114** prior to bonding may be controlled to intentionally form air gaps **172**. In other embodiments, air gaps **172** may be omitted.

(25) FIG. **8** illustrates the optional formation of additional structures in bonded integrated circuit structure **170**. For example, through substrate vias (TSVs, sometimes also referred to as through silicon vias or through vias) **174** may be formed in die **100**. TSVs **174** may be electrically connected to conductive features **106** in interconnect layers **104**. TSVs **174** may be formed using any suitable process. For example, a patterning process (e.g., photolithography and etching processes) may be used to pattern substrate **102** and/or one or more interconnect layers **104** to

expose conductive features **106**. Subsequently, barrier and/or a seed layers (not shown) may be deposited in the openings, and the openings may be filled with a conductive material to form TSVs **174** using a plating process, for example. Backside contact pads **176** may also be formed on TSVs **174**. In some embodiments, the formation of contact pads **176** may include using a hard mask to define a shape of contact pads **176**, depositing a seed layer, plating the contact pads, and removing the hard mask.

(26) Contact pads **176** may be used to electrically connect bonded integrated circuit structure **170** to other components. For example, connector elements, such as solder balls (not shown) may be used to connect contact pads **176** to other package components, such as, interposers, package substrates, fan-out redistribution layers (RDLs), and the like. As another example, contact pads **176** may be used to bond another integrated circuit die to bonded integrated circuit **170**. For example, FIG. **9** illustrates the optional bonding of another integrated circuit structure (die **180**) to die **100** using contact pads **176**. Die **180** may be similar to die **100**, and die **180** may include a substrate **192** and interconnect layers **190** having conductive features **194**. The bonding of dies **180** and **100** may use a similar process (e.g., hybrid bonding) as that described above for the bonding of die **100** to wafer **150**. In such embodiments, bonding layers **182** and **186** (e.g., oxide layers) may be used to bond die **180** to a surface of die **100** opposing wafer **150**. Furthermore, concave contact pads **184** may be used to bond to and contain the material of connectors **188**, and connectors **188**/contact pads **184** may electrically connect die **180** to die **100**. Additional features, such as TSVs **196** (e.g., electrically connected to conductive features **194**) and additional contact pads **198** may be formed in die **180**. Contact pads **198** may be used to electrically connect bonded integrated circuit structure to other package components, such as, additional dies, interposers, package substrates, fan-out RDLs, and the like. Additional processing may be performed, for example, to singulate portions of wafer **150** along scribe lines.

(27) FIG. **10** illustrates an example process **200** for forming a bonded integrated circuit structure in accordance with some embodiments. In step **202**, a first bonding layer (e.g., layer **108**) is formed over a first integrated circuit structure (e.g., die **100**), and a second bonding layer (e.g., layer **158**) is formed over a second integrated circuit structure (e.g., wafer **150**). In step **204**, the first and second bonding layers are patterned to include first openings (e.g., openings **110**) and second openings (e.g., openings **160**), respectively. The first openings may expose conductive features (e.g., conductive features **106'**) in the first integrated circuit structure, and the second openings may expose conductive features (e.g., conductive features **156'**) in the second integrated circuit structure.

(28) Next, in step **306**, convex connectors (e.g., connector **114**) are formed in the first openings. The connectors are referred to as convex because they may extend past a surface of the first bonding layer. The connectors are electrically connected to the exposed conductive features. In step **308**, concave contact pads (e.g., contact pads **164**) are formed in the second opening. The contact pads are referred to as concave because they cover sidewalls and bottom surfaces of the second openings while only partially filling the second openings. In step **310**, the first and second integrated circuit structures are bonded using a hybrid bonding process. The hybrid bonding process may include bonding the first and second bonding layers (e.g., the first and second bonding layers may be contacted together and may be annealed to form covalent bonds). The hybrid bonding process may further include bonding the convex connectors to the concave contact pads by disposing the convex connectors in remaining portions of the second openings. In some embodiments, the shape of the concave contact pads may advantageously contain the material of the connectors during annealing to prevent manufacturing defects, such as solder bridging, and the like.

(29) Various embodiments include a bonded integrated circuit structure, which may include two or more bonded integrated circuit structures (e.g., die-on-die bonding, wafer-on-wafer bonding, and/or die-on-wafer bonding). The bonding of the integrated circuit structures may include a hybrid

bonding process where bonding layers (e.g., oxide-to-oxide bonding layers) are disposed on each integrated circuit and bonded to other bonding layers using a bonding process, for example. Electrical connection between each integrated circuit structure may be achieved using conductive interconnect structures disposed between each integrated circuit structure. Such conductive interconnect structures include a concave contact pad (e.g., having a U-shape in a cross-sectional view), which may be used to contain the material of a convex connector (e.g., a solder or copper bump) during bonding. In the bonded structure, the concave contact pad may contact a bottom surface and sidewalls of the connector. Thus, the risk manufacturing defects, such as, solder bridging may be reduced, which allows for conductive interconnect structures to be spaced closer together (e.g., have a smaller pitch), increasing I/O density and improving yield.

(30) In accordance with an embodiment, a bonded integrated circuit (IC) structure includes a first IC structure and a second IC structure bonded to the first IC structure. The first IC structure includes a first bonding layer and a connector. The second IC structure includes a second bonding layer bonded to and contacting the first bonding layer and a contact pad in the second bonding layer. The connector extends past an interface between the first bonding layer and the second bonding layer, and the contact pad contacts a lateral surface and a sidewall of the connector.

(31) In accordance with another embodiment, a method includes patterning a first opening in a first bonding layer of a first IC structure and patterning a second opening in a second bonding layer of a second IC structure. The method further includes forming a connector in the first opening and forming a contact pad on sidewalls and a bottom surface of the second opening. The connector extends past a lateral surface of the first bonding layer. The first and the second IC structures are bonded. Bonding the first and the second IC structures includes fusion bonding the first bonding layer to the second bonding layer and bonding the connector to the contact pad. The connector is partially disposed in the second opening.

(32) In accordance with yet another embodiment, a bonded integrated circuit (IC) structure includes a first semiconductor substrate, a first bonding layer over the first semiconductor substrate, and a contact pad in the first bonding layer. The bonded IC structure further includes a second bonding layer over and contacting the first bonding layer, a connector disposed in the first bonding layer and the second bonding layer, and a second semiconductor substrate over the second bonding layer. The contact pad is disposed on a lateral surface and a sidewall of the connector.

(33) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method for forming a semiconductor device, the method comprising: forming, in a first semiconductor structure, a first bonding layer comprising a first dielectric layer and a plurality of protruding contact structures; forming, in a second semiconductor structure, a second bonding layer comprising a second dielectric layer and a plurality of recess contact structures, wherein each of the plurality of recess contact structures comprises a metal feature, a surface of the metal feature of each of the plurality of recess contact structures having a first recess; and after forming the first bonding layer and after forming the second bonding layer, bonding the plurality of protruding contact structures with the plurality of recess contact structures such that each of the plurality of protruding contact structures is in contact with the first recess of the surface of a respective recess

contact structure.

2. The method of claim 1, wherein forming the second bonding layer comprises: forming a second recess in the second dielectric layer; forming a conductive layer over the second dielectric layer and in the second recess; and performing a planarization to remove portions of the conductive layer over the second dielectric layer.
3. The method of claim 2, wherein the second recess extends completely through the second dielectric layer.
4. The method of claim 1, wherein the plurality of the recess contact structures comprise copper, nickel, gold, tin, silver, aluminum, alloys thereof, or combinations thereof.
5. The method of claim 1, wherein bonding deforms the plurality of protruding contact structures.
6. The method of claim 1, wherein, after bonding, the first bonding layer completely covers an upper surface of the plurality of recess contact structures.
7. A method for forming a semiconductor device, the method comprising: forming a first dielectric layer over a first substrate; patterning a first recess in the first dielectric layer; forming a contact pad in the first recess; forming a second dielectric layer over a second substrate; patterning a second recess in a second dielectric layer; forming a connector in the second recess, wherein the connector extends past a first surface of the second dielectric layer; and after the forming the contact pad and after the forming the connector, bonding the first substrate and the second substrate, wherein the first surface of the second dielectric layer faces the first substrate, wherein bonding the first substrate and the second substrate comprises bonding the connector to the contact pad, wherein the connector is partially disposed in the second recess.
8. The method of claim 7, wherein a surface of the contact pad is level with the first surface of the first dielectric layer.
9. The method of claim 7, wherein bonding the first substrate to the second substrate comprises bonding the first dielectric layer to the second dielectric layer.
10. The method of claim 7, wherein bonding the first substrate to the second substrate reshapes the connector.
11. The method of claim 7, wherein after bonding, the connector contacts a surface of the contact pad at a bottom of the first recess.
12. The method of claim 11, wherein after bonding, the connector contacts a portion of the contact pad that is in contact with a sidewall of the first recess.
13. The method of claim 7, wherein the second recess extends completely through the second dielectric layer, and wherein the first recess extends completely through the first dielectric layer.
14. The method of claim 7, wherein a width of the first recess is greater than a width of the second recess.
15. A method comprising: forming a first bonding layer on a first semiconductor structure; forming a first recess in the first bonding layer, the first recess exposing a first conductive feature of the first semiconductor structure; forming a first contact in the first recess, the first contact comprising a protrusion that protrudes from the first bonding layer; forming a second bonding layer on a second semiconductor structure; forming a second recess in the second bonding layer, the second recess exposing a second conductive feature of the second semiconductor structure; forming a second contact in the second recess, an upper surface of the second contact being below an upper surface of the second bonding layer, wherein the second contact has a U-shape in a cross-sectional view; and after forming the first contact and after forming the second contact, bonding the protrusion of the first contact to the upper surface of the second contact, wherein after bonding, the protrusion extends into the second recess.
16. The method of claim 15, wherein bonding comprises bonding the first bonding layer to the second bonding layer, wherein, after bonding, a width of the protrusion level with an interface between the first bonding layer and the second bonding layer is less than a width of the protrusion in the second recess.

17. The method of claim 15, wherein bonding further comprises bonding the first bonding layer to the second bonding layer.
 18. The method of claim 15, wherein, after bonding, the second contact directly contacts the first bonding layer.
 19. The method of claim 15, wherein, after bonding, the protrusion and the second contact completely fill a bottom of the second recess.
 20. The method of claim 15, wherein, after bonding, the protrusion extends under the second bonding layer such that the protrusion is between the first semiconductor structure and the second contact along a line perpendicular to a major surface of the first bonding layer.
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